

1. (a) What set is the union of $\{8, 16, 22\}$ and the intersection of $\{4, 5, 6, 14, 15, 16, 24, 25, 26\}$ and $\{32, 24, 16, 8\}$?

Answer:

$$8, 16, 22 \cup 4, 5, 6, 14, 15, 16, 24, 25, 26 \cap 32, 24, 16, 8$$

- (b) What set is the intersection of $\{10, 30, 20\}$ and the union of $\{10, 20\}$ and $\{20, 40\}$?

- (c) What is $\{3, 5, 7, 9\} - \{7, 6, 5, 4\}$?

- (d) What is $\{4, 6, 5, 7\} - \{3, 5, 7, 9\}$?

- (e) Let $A = \{100, 200, 4, 2, -8\}$. Let $B = \{0, 4, -8, 100, 200\}$.
What is $(A - B) - (B - A)$?

- (f) Let $A = \{100, 200, 4, 2, -8\}$. Let $B = \{0, 4, -8, 100, 200\}$
What is $A - (A - B)$?

- (g) Use set builder notation to express $\{3, 7, 11, 15, \dots, 403\}$, as compactly as possible, with as little English as possible.

2. Consider the following sets: $H = \{\text{herbivores}\}$. $S = \{\text{animals with sharp teeth}\}$. $R = \{\text{animals that run fast}\}$. $L = \{\text{animals with long necks}\}$.

(a) Use set notation (e.g., union, intersection, etc) to describe herbivores that have sharp teeth or run fast. Your expression should be as simple as possible. *Do not use set builder notation.*

(b) To survive the next ice age, members of a species must have sharp teeth or run fast, but they must also have long necks or have sharp teeth. Express these conditions using set notation, minimizing redundancy.

(c) In a certain herbivore ecosystem, there are 52 species that have sharp teeth or can run fast. 18 of these species have both qualities. 25 species have sharp teeth. How many of the 52 species can run fast? (Briefly show how you calculated this, and name the most relevant concept in the calculation)

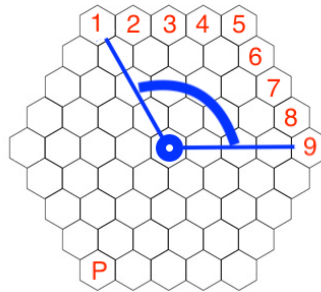
It is ok to use the symbol of a set to denote its size, if context is clear; no need to use $|$ $|$.

(d) The ecosystem mentioned in (c) contains 100 species.

i. How many don't have sharp teeth and can't run fast? Use deMorgan's laws.

ii. How many don't have sharp teeth or can't run fast (or both)? Use deMorgan.

5. Suppose that we play a variant of *Chomp* on a hexagonal board like the one below. The poisoned cookie is marked as **P**. Whenever you take a cookie, you must also take everything within its “upper-right” 120° cone. For example, if you take the middle cookie in the illustration then you must take cookies 1 through 9 and everything else in the marked region.



Prove that Player 1 can win this game, with a non-constructive proof.